

AirFare

MVP Build Specification

Version 1.0 · Frontend, API contracts, database, configuration

This document is the implementation layer. The PRD says what AirFare is for and the TRD says how the system is architected. This document says what to type. It is written so that a developer who has read neither of the other two can build the correct screen, endpoint or table.

Field	Detail
Document	MVP Build Specification v1.0
Companions	AirFare PRD v2.0 · AirFare TRD v2.0
Covers	Design system, page specs, chart specs, API contracts, database schema, configuration, seed data, acceptance criteria
Does not cover	Scraper adapter internals and statistical method derivations. Those live in the TRD.
Audience	Whoever is writing the code, on day one, without further explanation

Contents

0. How To Use This Document

If you are	Read
Building the frontend	Parts A, B, C, D. Build against the mock contracts in Part E; you do not need the backend to start.
Building the API	Parts E, F, G. Every endpoint has its full request and response shape.
Building the pipeline	The TRD, plus Part F here for the exact schema.
Reviewing progress	Part I. Every screen has acceptance criteria; a screen is either done or it is not.

The one rule that matters most

The frontend is built against the contracts in Part E and the seed data in Part H from day one. It does not wait for scrapers. If frontend work is blocked on collection working, the build window is lost to serialised effort and the dashboard will be unfinished at demonstration.

1. Design Tokens

Define these once as CSS custom properties. Nothing in the application uses a raw hex value.

1.1 Surfaces and ink

Token	Light	Dark	Use
--page	#f9f9f7	#0d0d0d	Application background
--surface	#fcfcfb	#1a1a19	Cards, panels, chart surfaces
--surface-raised	ffffff	#242423	Modals, dropdowns, tooltips
--brand	#12304a	#12304a	Top bar, primary nav. Chrome only, never a data encoding.
--brand-accent	#0b6e8f	#2a9ab8	Links, active nav, focus rings
--text-primary	#0b0b0b	ffffff	Headings, values
--text-secondary	#52514e	#c3c2b7	Body, descriptions
--text-muted	#898781	#898781	Axis labels, captions, metadata
--grid	#e1e0d9	#2c2c2a	Chart gridlines, hairline dividers
--axis	#c3c2b7	#383835	Chart baseline and axis lines
--border	rgba(11,11,11,.10)	rgba(255,255,255,.10)	Card and input borders

1.2 Type scale

One family throughout: system-ui, -apple-system, "Segoe UI", sans-serif. No display or serif face anywhere.

Token	Size	Weight	Use
--t-hero	48px	600	The index value on the Overview page. Proportional figures.
--t-h1	28px	600	Page titles
--t-h2	20px	600	Card and section titles
--t-body	14px	400	Default body text
--t-small	13px	400	Captions, metadata, axis labels
--t-micro	11px	500	Badges, chip labels. Uppercase, 0.04em tracking.

Tabular figures (font-variant-numeric: tabular-nums) are applied only to table columns and axis ticks, where digits must align vertically. The hero number and stat-tile values use default proportional figures.

1.3 Spacing, radius, elevation

```

space    4 · 8 · 12 · 16 · 24 · 32 · 48 · 64      (4px base)
radius   --r-sm 4px  --r-md 8px  --r-lg 12px  --r-pill 999px
shadow    --e-1  0 1px 2px rgba(11,11,11,.06)  cards
          --e-2  0 4px 12px rgba(11,11,11,.10)  dropdowns, tooltips
          --e-3  0 12px 32px rgba(11,11,11,.16)  modals
motion    --dur-fast 120ms  --dur-base 200ms  ease-out
          respect prefers-reduced-motion: no transform or opacity transitions

```

2. Chart Palette

The chart palette is separate from the brand palette. Brand navy is chrome. Series colours carry data and are therefore chosen for perceptual separation rather than for brand fit.

2.1 Categorical series

Assigned in fixed slot order and never cycled. Colour follows the entity, so filtering out a carrier must not repaint the remaining ones.

Slot	Hue	Light	Dark	Assigned to
1	blue	#2a78d6	#3987e5	AirFare headline / IndiGo
2	orange	#eb6834	#d95926	TCT-AirFare / Air India
3	aqua	#1baf7a	#199e70	AirFare uncorrected / Akasa Air
4	yellow	#eda100	#c98500	ANCx / SpiceJet
5	magenta	#e87ba4	#d55181	MakeMyTrip
6	green	#008300	#008300	Reserve
7	violet	#4a3aa7	#9085e9	Reserve
8	red	#e34948	#e66767	Reserve

Validation result

Validated against the light chart surface #fcfcfb with all pairs in play, which is the strict case needed for scatter and small multiples:

Lightness band	PASS	all inside L 0.43-0.77
Chroma floor	PASS	all >= 0.1
CVD separation	PASS	worst all-pairs dE 9.2 (deutan)
Normal-vision floor	PASS	worst all-pairs dE 24.0
Contrast vs surface	WARN	aqua at 2.74:1

The contrast warning obliges relief, so every chart using slot 3 carries visible direct labels or an accessible table view. This is not dismissable. Beyond three simultaneous series in a scatter or small-multiple context, fold the remainder into "Other" or facet, because yellow beside orange fails the all-pairs floor.

2.2 Sequential ramp

One hue, light to dark, used for the lead-time heatmap and any magnitude encoding. Blue.

100 #cde2fb	250 #86b6ef	400 #3987e5	550 #1c5cab
150 #b7d3f6	300 #6da7ec	450 #2a78d6	600 #184f95
200 #9ec5f4	350 #5598e7	500 #256abf	650 #104281
			700 #0d366b

2.3 Diverging ramp

Blue to red with a neutral grey midpoint (light #f0efec, dark #383835). Used only where a value has a genuine zero crossing: the attribution waterfall and period-on-period change. Equal step count per arm. Never a hue at the midpoint.

2.4 Status palette

Reserved. Never reused as a series colour. Always shipped with an icon and a label, so state is never carried by colour alone.

Role	Hex	Use in AirFare
good	#0ca30c	Source HEALTHY, quality score above target
warning	#fab219	Source DEGRADED, imputation rate above 15%
serious	#ec835a	Coverage below 90%, provisional value
critical	#d03b3b	Circuit OPEN, index publish failed

2.5 Texture

A single directional line fill at 45 degrees, mirrored at 135 degrees, tone-on-tone. Applied to simulated-provenance regions and triggered by the accessibility setting, print, or forced-colors. Never decorative and never on by default.

3. Component Inventory

Build these before building pages. Every page is assembled from this list.

Component	Specification
AppShell	Top bar with brand, environment badge, user menu. Left nav, collapsible below 1024px. Content area with a max width of 1440px and a 24px gutter.
NavItem	Icon plus label. Active state uses a 3px left rule in --brand-accent plus a background wash. Role-gated items are not rendered at all rather than disabled.
PageHeader	Title, optional subtitle, a right-aligned action slot, and a date-context line stating which collection run the page reflects.
Card	Surface, 1px border, --r-md, --e-1, 20px padding. Optional header with title and a right slot for controls.
StatTile	Label, value, delta with direction arrow, optional sparkline. Delta uses the success and critical status colours with an arrow glyph, never colour alone.
HeroValue	The 48px index number with its base, as-of date, and a QualityBadge. Used once per page at most.
QualityBadge	Pill showing coverage percent and a status dot. Click opens the quality detail popover. Appears next to every published value without exception.
ProvenanceTag	REAL or SIMULATED. Simulated renders with the texture fill and is never visually quieter than the real tag.
VintageTag	Shows vintage number and provisional state. Only rendered when vintage is greater than 1 or the value is provisional.
SourceStatusDot	Status colour dot plus text label plus icon. Four states: HEALTHY, DEGRADED, OPEN, RECOVERING.
DateRangePicker	Preset rows: 7 days, 30 days, 90 days, month to date, custom. Selection marked with a 16px bold check. Custom range behind a hairline in the footer.
FilterBar	One horizontal row above the charts. Holds route, carrier, source, window and series selectors. Never a sidebar.
ChartFrame	Wraps every chart. Provides title, subtitle, legend slot, a table-view toggle, a download control and the five states from Section 5.
DataTable	Sortable, sticky header, tabular figures, zebra rows in --alt. Serves as the accessible alternative for every chart.
Tooltip	Surface-raised, --e-2, 8px padding, --t-small. Never covers the mark it describes.
Drawer	Right-side panel for observation detail. Closes on Escape and on backdrop click.
EmptyState	Icon, one line of explanation, one action. Distinct from an error state in both wording and colour.
Skeleton	Shape-matched loading placeholder. Never a spinner over a whole page.

Component	Specification
Toast	Transient notification, top right, auto-dismiss at 5s, status-coloured with an icon.

4. Layout and Responsive Behaviour

4.1 Grid

Content max-width 1440px, centred, 24px gutter
Grid 12 columns, 24px gap
Card spans full 12 · half 6 · third 4 · quarter 3
Vertical rhythm 24px between cards, 32px between sections

4.2 Breakpoints

Name	Width	Behaviour
xl	>= 1440px	Full layout. Four stat tiles per row, charts at half width.
lg	1024-1439px	Nav collapses to icons. Three stat tiles per row.
md	768-1023px	Nav becomes a drawer. Two stat tiles per row. Charts go full width.
sm	< 768px	Single column. Heatmap becomes horizontally scrollable with a sticky first column. Tables become stacked key-value cards.

The analyst dashboard is desktop-first because that is where it is used. It must remain legible on a tablet, because that is what someone will be holding at a demonstration. Phone support is a courtesy, not a target.

5. Component States

Every data-bearing component implements five states. Most dashboards implement two and break in front of an audience.

State	Behaviour
Loading	Shape-matched skeleton with the real axis frame drawn. No spinner over the page, no layout shift when data arrives.
Empty	Data loaded successfully and there is legitimately nothing. Neutral colour, explains why in one line, offers one action. Example: no observations for this route on this date.
Error	The request failed. Critical status colour, states what failed, offers retry, shows a correlation id for the log. Never silently shows zero.
Partial	Data present but below quality threshold. Renders normally with a serious-status band across the top of the card naming the issue. Never hidden and never presented as clean.
Stale	The most recent collection run is older than expected. Renders with a muted overlay and the age of the newest observation.

5.1 The data-quality states, which are specific to this product

Because AirFare publishes numbers that may be imputed, provisional or simulated, the interface must make that visible at the point of reading, not in a footnote. These rules are not optional decoration; they are the honesty commitment from the PRD made visible.

Condition	Required visual treatment
Imputation rate > 15%	Warning band on the card. The imputation percentage is shown beside the value, not only in the quality page.

Condition	Required visual treatment
Coverage < 90%	Serious band on the card. The QualityBadge turns serious. The value still renders.
is_provisional true	The value carries a "Provisional" chip. Chart points in the provisional region use a hollow marker.
provenance SIMULATED	The chart region is filled with the 45-degree texture, the legend carries a SIMULATED entry, and the boundary between real and simulated is drawn as a labelled vertical rule. Never inferred from date; always read from the field.
vintage > 1	A revision indicator sits beside the value and links to the vintage history.
Cell flagged LOW_CONFIDENCE	Heatmap cell is hatched rather than filled. Tooltip states the observation count.

The rule behind all of this

A viewer must never be able to mistake a modelled, imputed or simulated number for a measured one. If a treatment above is ambiguous at a glance, it is wrong, and the correct fix is more contrast, not a tooltip.

PART B · INFORMATION ARCHITECTURE

6. Route Map

Route	Access	Page
/login	public	Sign in
/	VIEWER+	Overview
/attribution	ANALYST+	Movement attribution
/cost-comparison	VIEWER+	Base fare versus total cost
/routes	VIEWER+	Route list
/routes/[routeId]	VIEWER+	Route detail
/lead-time	VIEWER+	Lead-time heatmap
/carriers	VIEWER+	Carrier and OTA comparison
/components	ANALYST+	Fare component breakdown
/quality	VIEWER+	Data quality
/health	ANALYST+	Collection health
/methodology	public	Methodology and versions
/observations/[id]	ANALYST+	Single observation detail (drawer or page)
/public	public	Citizen transparency view (Tier 2)
/admin	ADMIN	Source policy registry, weights, users (Tier 2)

6.1 Navigation grouping

INDEX	Overview · Attribution · Cost comparison
EXPLORE	Routes · Lead time · Carriers · Components
TRUST	Quality · Health · Methodology
ADMIN	Admin (ADMIN role only, hidden otherwise)

The third group is named TRUST deliberately. Grouping quality, collection health and methodology together makes the auditability argument visible in the navigation itself, which is a point in your favour that costs nothing to make.

7. Authentication and Roles

7.1 Flow

```
POST /auth/login {email, password}
-> 200 {access_token, refresh_token, expires_in, user{id,name,role}}
-> 401 invalid credentials
```

Client stores the access token in memory and the refresh token in an httpOnly cookie. Every request sends: Authorization: Bearer <token>

On 401 mid-session: attempt one silent refresh, then redirect to /login preserving the intended destination.

7.2 Roles

Role	Sees	Cannot
VIEWER	Overview, cost comparison, routes, lead time, carriers, quality, methodology	Attribution, components, health, admin, observation detail
ANALYST	Everything a viewer sees, plus attribution, components, health, observation detail, CSV and SDMX export	Admin
ADMIN	Everything, plus source policy registry, weight management, user management	
SERVICE	API only, no interface	Any write endpoint

Role-gated navigation items are not rendered, not disabled. A disabled item advertises the existence of something the user cannot have and invites a support question.

PART C · PAGE SPECIFICATIONS

8. Pages

Each page below gives its route, access, purpose, layout, data dependencies, components and acceptance criteria. A page is complete when every acceptance criterion passes, not when it looks finished.

8.1 Overview

Field	Specification
Route	/
Access	VIEWER and above
Purpose	The single screen that answers "what is airfare inflation today, and can I trust today's number". This is the demonstration landing page and receives the most design attention.

Field	Specification
Layout	Row 1: HeroValue (span 5) beside a 2x2 StatTile grid (span 7). Row 2: main index time series, full width, 420px tall. Row 3: source health strip (span 4) and top movers list (span 8).
Data sources	GET /index/latest · GET /index/history · GET /quality/coverage · GET /sources
Components	HeroValue, QualityBadge, ProvenanceTag, StatTile x4, ChartFrame with multi-series line, SourceStatusDot strip, DataTable for top movers

The four stat tiles are: daily change, weekly change, monthly change, and coverage. Coverage is a stat tile rather than a footnote because quality disclosure is a product commitment and putting it fourth in the primary row is what makes that commitment real.

Acceptance criteria

- The hero value renders with its base period, as-of date and a quality badge, and none of the four can be absent.
- A value with coverage below 90 per cent renders with the serious band and still shows the number.
- The simulated region of the time series is visibly textured and labelled in the legend, and the real-to-simulated boundary is drawn as a labelled rule.
- Loading shows shape-matched skeletons with the axis frame already drawn and no layout shift on arrival.
- A failed request shows the error state with a retry control and a correlation id, and never renders zero.
- The chart is keyboard navigable and the table view toggle produces an equivalent DataTable.
- The page is legible and complete at 768px width.

8.2 Attribution

Field	Specification
Route	/attribution
Access	ANALYST and above
Purpose	Explains why the index moved. This is PRD USP 4 and the strongest single demonstration view in the product.
Layout	Row 1: date selector and total movement summary. Row 2: waterfall chart, full width, 460px tall. Row 3: four small contribution tables side by side (route, carrier, window, fee category).
Data sources	GET /index/attribution/{date}
Components	DateRangePicker (single date mode), ChartFrame with waterfall, DataTable x4, Tooltip

The waterfall opens at the previous value, shows each contribution as a floating bar using the diverging ramp, and closes at the current value. The sum of contributions must equal the total movement and the frontend asserts this, showing a reconciliation warning if the API returns a set that does not sum.

Acceptance criteria

- Contributions sum to the stated total movement, and a mismatch renders a visible warning rather than being silently absorbed.
- Positive and negative contributions use the diverging pair with a neutral zero line, and no hue sits at the midpoint.
- Hovering a bar shows the dimension, its contribution in index points, and its percentage of the total.
- Clicking a route contribution navigates to that route detail page.
- A date with no attribution data shows the empty state, not a zero waterfall.
- The table view lists every contribution with its value.

8.3 Base Fare versus Total Cost

Field	Specification
Route	/cost-comparison
Access	VIEWER and above
Purpose	Shows AirFare headline and TCT-AirFare on one chart. The gap between them is the measured blind spot and is the core evidence in the PRD argument.
Layout	Row 1: headline gap figure as a HeroValue with its own explanation line. Row 2: dual-series line chart with the gap shaded, full width. Row 3: fee composition stacked bar showing what makes up the gap.
Data sources	GET /index/family · GET /index/history?series=AirFare,TCT-AirFare
Components	HeroValue, ChartFrame with line plus shaded band, ChartFrame with stacked bar, DataTable

Both series are on one axis because both are index values on the same base. This is the one place a dual axis would be tempting and it must not be used; if the two ever need different scales, that is a signal the base period is wrong.

Acceptance criteria

- Both series share a single y-axis. No dual-axis chart exists anywhere in the application.
- The gap between the two series is shaded and the current gap is stated numerically in the hero.
- Both series are direct-labelled at their right edge in addition to the legend.
- The stacked bar segments carry a 2px surface gap between them.
- Aqua-slot series carry visible direct labels, satisfying the contrast relief requirement.

8.4 Route List and Route Detail

Field	Specification
Route	/routes and /routes/{routeId}
Access	VIEWER and above
Purpose	List: compare all basket routes at a glance. Detail: drill from a route index down to the individual observations behind it, which is the reproducibility guarantee made usable.
Layout	List: filter bar, then a sortable table with sparkline per row. Detail: route header with weight and state, index time series, window breakdown chart, and an observations table with a drawer.
Data sources	GET /routes · GET /routes/{id}/index · GET /routes/{id}/fares · GET /observations
Components	FilterBar, DataTable with inline sparklines, ChartFrame, Drawer, QualityBadge per row

The observations table is paginated and each row opens the Drawer showing the full canonical observation including adapter version, parser version, fetch tier and the raw artefact reference. That drawer is the visible proof of provenance.

Acceptance criteria

- The route list is sortable by index value, daily change, coverage and weight.
- Each row shows its normalised weight, and the weights visibly sum to 100 per cent.
- Route detail reaches an individual observation in at most three clicks from the national index.

- The observation drawer shows adapter version, parser version, fetch tier, provenance and the raw artefact reference.
- A route with an open circuit breaker shows the critical status and its last successful observation time.

8.5 Lead-Time Heatmap

Field	Specification
Route	/lead-time
Access	VIEWER and above
Purpose	Route by advance-purchase window. The single most legible visual for a non-technical audience, because the effect of booking early is immediately visible without explanation.
Layout	Filter bar with date and carrier. Heatmap with routes as rows and windows as columns, cell labels on. Below, a lead-time elasticity curve per selected route.
Data sources	GET /routes/{id}/fares grouped by window, or a dedicated GET /lead-time/matrix
Components	FilterBar, ChartFrame with heatmap, ChartFrame with line, DataTable

Cells are coloured on the sequential blue ramp by fare level and labelled with the rupee value. Low-confidence cells are hatched rather than filled, with the observation count in the tooltip.

Acceptance criteria

- Every cell carries its value as a visible label, so colour is never the only encoding.
- The colour scale legend states its range and is anchored consistently across dates, so two dates are comparable.
- Low-confidence and imputed cells are visually distinct from measured cells.
- On screens below 768px the heatmap scrolls horizontally with a sticky route column.
- A cell with no observation is rendered as an explicit no-data cell, never as the lightest colour in the ramp.

8.6 Carrier and OTA Comparison

Field	Specification
Route	/carriers
Access	VIEWER and above
Purpose	Shows the price difference between booking direct with a carrier and booking through an aggregator, including the platform fee wedge that exists nowhere else in public data.
Layout	Row 1: stat tiles for average direct fare, average aggregator fare and the mean wedge. Row 2: grouped bar by route. Row 3: fee wedge trend over time.
Data sources	GET /carriers · GET /sources · GET /index/history?group=source_type
Components	StatTile x3, ChartFrame with grouped bar, ChartFrame with line, DataTable

Acceptance criteria

- Carrier colours are assigned from the fixed slot order and do not change when the filter reduces the series count.
- Adjacent bars carry a 2px surface gap.
- The wedge is stated as both an absolute rupee figure and a percentage.
- Where a carrier is not sampled on a route, the bar is absent with a note, not drawn at zero.

8.7 Fare Components

Field	Specification
Route	/components
Access	ANALYST and above
Purpose	Decomposes the total payable amount into base fare, surcharges, statutory fees, tax, platform fee and ancillaries, and shows how that mix has shifted.
Layout	Row 1: current composition as a stacked bar, one bar per route. Row 2: composition over time as a stacked area. Row 3: component table with completeness per component.
Data sources	GET /observations aggregated, or a dedicated GET /components/breakdown
Components	ChartFrame with stacked bar, ChartFrame with stacked area, DataTable

Components that the source does not expose are null, not zero. The chart must render these as an explicit unknown segment with the hatched texture, because drawing them as zero would misrepresent the decomposition.

Acceptance criteria

- Null components render as an explicit unknown segment, never as zero and never omitted.
- The component table shows completeness per component so a viewer can see how much of the decomposition is actually observed.
- Stacked segments carry a 2px surface gap.
- Segment order is fixed across all bars so the eye can compare across routes.

8.8 Data Quality

Field	Specification
Route	/quality
Access	VIEWER and above
Purpose	The trust page. Shows coverage, completeness, imputation rate, source agreement, freshness and provenance mix, so a user can judge how much weight to put on today's number.
Layout	Row 1: overall quality score as a HeroValue with five component stat tiles. Row 2: coverage over time. Row 3: imputation breakdown by rule. Row 4: source agreement matrix.
Data sources	GET /quality/coverage · GET /quality/sources · GET /quality/imputation
Components	HeroValue, StatTile x5, ChartFrame x3, DataTable

This page exists because the PRD commits to disclosing uncertainty on the face of the output. It should look as considered as the Overview page, not like a diagnostics screen.

Acceptance criteria

- All five component scores render with their own targets and a pass or fail state.
- The imputation breakdown shows counts by rule, so a viewer can see whether gaps were filled from another source or carried forward.
- Provenance mix shows the real versus simulated split for the selected period.
- Any metric below target uses the status palette with an icon and a label, never colour alone.

8.9 Collection Health

Field	Specification
Route	/health
Access	ANALYST and above
Purpose	Operational view of every source adapter. Distinct from Grafana, which is for engineers; this is for an analyst who needs to know whether today's data can be trusted.
Layout	Row 1: status strip, one card per source. Row 2: success rate over time. Row 3: recent run table with per-run job counts and failures.
Data sources	GET /sources · GET /quality/sources
Components	SourceStatusDot cards, ChartFrame with line, DataTable

Acceptance criteria

- All four circuit states render distinctly with icon, colour and text label.
- Each source shows its last successful observation timestamp and its current rate limit.
- A source in the BLOCKED state is shown as blocked with its policy record, and the interface offers no action that would retry or escalate.
- The recent runs table links each run to the observations it produced.

8.10 Methodology

Field	Specification
Route	/methodology
Access	Public
Purpose	Publishes the formulas, weights, model versions, base period and revision history. Open methodology is a stated product commitment, so it needs a page rather than a document.
Layout	Single column, readable width. Sections: index formula, current weights table, elementary aggregate, quality adjustment model version, booking curve version and calibration source, base period, revision log.
Data sources	GET /methodology · GET /index/vintages/{date}
Components	Static content plus DataTable for weights and revisions

Public access is deliberate. A methodology page behind a login contradicts the transparency claim the whole product rests on.

Acceptance criteria

- Current weight table renders with raw and normalised weights, and the normalised column visibly sums to 1.
- Every model version in force is named with its estimation date.
- The revision log lists every revision with its prior value and reason.
- The page is reachable and readable without authentication.

8.11 Login

Field	Specification
Route	/login

Field	Specification
Access	Public
Purpose	Sign in. Deliberately minimal.
Layout	Centred card, 400px, on the page plane. Brand mark, email, password, submit, error slot.
Data sources	POST /auth/login
Components	Card, form inputs, Toast

Acceptance criteria

- Invalid credentials show an inline error that does not reveal whether the email exists.
- A successful login redirects to the originally intended route when there was one.
- The form is submittable by keyboard alone and the first field is focused on mount.

PART D · CHART SPECIFICATIONS

9. Chart Specifications

Chart form is chosen by the job the data does, not by preference. Each chart below states its form, encoding, colour role, interaction and empty behaviour.

Chart	Form	Encoding	Colour role	Page
Index time series	Multi-line	x: date, y: index value	Categorical, slots 1-3	Overview
Attribution	Waterfall	x: dimension, y: index points	Diverging, zero midpoint	Attribution
Base vs total	Line + band	x: date, y: index, band: gap	Categorical slots 1-2	Cost comparison
Fee composition	Stacked bar	x: route, y: rupees	Categorical, fixed order	Cost comparison
Lead-time matrix	Heatmap	x: window, y: route, fill: fare	Sequential blue	Lead time
Elasticity curve	Line	x: advance days, y: fare	Categorical slot 1	Lead time
Carrier compare	Grouped bar	x: route, y: fare, group: carrier	Categorical, fixed order	Carriers
Component mix	Stacked area	x: date, y: share	Categorical, fixed order	Components
Coverage trend	Line	x: date, y: percent	Categorical slot 1	Quality
Source agreement	Heatmap	x: source, y: source, fill: dispersion	Sequential blue	Quality
Route sparkline	Sparkline	x: date, y: index	Text-secondary ink	Route list

9.1 Universal mark rules

- Lines are 2px. Markers are at least 8px. Bar data-ends are 4px rounded and anchored to the baseline.
- A 2px surface gap sits between stacked segments and between adjacent bars.
- Overlapping marks carry a 2px surface ring.
- Gridlines are hairline in --grid and recede. Axes in --axis. Neither competes with the data.
- Direct labels are selective. Never a number on every point; label the first, last and any annotated extreme.
- Text always wears text tokens, never the series colour. A coloured mark beside the label carries identity.

9.2 Universal interaction rules

- Line and area charts get a crosshair with a shared tooltip listing every series at that x position.
- Bar, dot and heatmap cells get a per-mark hover tooltip. Hit targets are larger than the mark.
- Filters sit in one horizontal row above the charts, never in a sidebar.
- Every chart has a table-view toggle producing an equivalent DataTable. This is the accessible alternative and also the contrast relief for slot 3.
- Tooltips never cover the mark they describe.

9.3 Rules that must not be broken

Rule	Why
No dual-axis charts, anywhere	Two y-scales on one plot is the most common serious chart error and makes any correlation look arbitrary. Two measures of different scale become two charts or are indexed to a common base.
Categorical hues in fixed order, never cycled	A ninth series folds into Other or facets. A generated hue breaks the validated separation.
Colour follows the entity, never the rank	Filtering out a carrier must not repaint the survivors, or the reader loses the thread across views.
Sequential is one hue light to dark	A rainbow ramp implies category boundaries that do not exist in a continuous measure.
Diverging has a neutral grey midpoint	A hue at the midpoint reads as a third category rather than as zero.
Status colours are never series colours	A series that borrows the critical red will be read as an alert.
Legend present for two or more series	Identity is never colour alone. Four or fewer series are also direct-labelled.
Dark mode is selected, not flipped	Dark values come from the dark column of the palette and are validated against the dark surface.

PART E · API CONTRACTS

10. Conventions

Base URL	/api/v1
Auth	Authorization: Bearer <jwt>
Content type	application/json; charset=utf-8
Dates	ISO 8601 date (YYYY-MM-DD) or datetime with Z
Money	integer paise, plus a formatted string, never a float
Nulls	an unknown value is null; it is never 0 and never omitted

10.1 Envelope

Every successful list response uses the same envelope so the client has one parser.

```
{
  "data": [ ... ],
  "meta": {
    "page": 1, "page_size": 50, "total": 1284,
    "generated_at": "2026-09-21T02:47:11Z"
  }
}
```

10.2 Error shape

```
{
  "error": {
    "code":
"ROUTE_NOT_FOUND",

    "message":
"No route
with id
DEL-XXX in
the active
basket.",

    "correlation_
id":
"c1f3e9a2-...",
    "details": {}
  }
}
```

Status	Meaning
400	Malformed request or invalid parameter value
401	Missing, expired or invalid token
403	Authenticated but the role does not permit this resource
404	Resource does not exist
409	Conflict, for example requesting a vintage that has been superseded
422	Well-formed but semantically invalid, for example a date before the base period
429	Rate limit exceeded. Retry-After header is set.
503	Index not yet published for the requested date

10.3 Quality metadata, mandatory on every index response

No endpoint returns an index value without these fields. The client cannot render a number without also being able to render how good it is, which is enforced at the contract level rather than by convention.

```
"coverage": 0.96,
"imputation_rate": 0.04,
"provenance": "REAL" | "SIMULATED" | "MIXED",
"vintage": 1,
"is_provisional": false,
"methodology_version": "2.0.1",
"quality_status": "GOOD" | "WARNING" | "SERIOUS" | "CRITICAL"
```

11. Endpoint Contracts

11.1 GET /index/latest

```
Query    series  AirFare | AirFare-uncorrected | TCT-AirFare | ANCx   (default AirFare)
Role     VIEWER

200
{
  "series": "AirFare",
  "date": "2026-09-20",
  "value": 108.42,
  "base": 100,
  "base_period": "2026-09-01/2026-09-30",
  "change": { "daily": 0.74, "weekly": 2.18, "monthly": 4.06 },
  "coverage": 0.96, "imputation_rate": 0.04,
  "provenance": "REAL", "vintage": 1, "is_provisional": false,
  "methodology_version": "2.0.1", "quality_status": "GOOD",
```

```
"calculated_at": "2026-09-21T02:47:11Z"
}
```

503 index not yet published for the latest collection date

11.2 GET /index/history

```
Query  series      repeatable, default AirFare
      from, to    ISO date, default last 30 days
      frequency  daily | weekly | monthly (default daily)
      seasonally_adjusted boolean, default false
Role   VIEWER

200
{
  "data": [
    { "date": "2026-09-19", "series": "AirFare", "value": 107.66,
      "coverage": 0.97, "imputation_rate": 0.03,
      "provenance": "REAL", "is_provisional": false }
  ],
  "meta": { "series": ["AirFare"], "frequency": "daily",
    "base_period": "2026-09-01/2026-09-30",
    "provenance_boundary": "2026-09-09" }
}
```

provenance_boundary is the date at which the series changes from simulated to real. The chart draws its labelled rule at this date and must not infer it from anything else.

11.3 GET /index/family

```
Query  date      ISO date, default latest
Role   VIEWER

200
{
  "date": "2026-09-20",
  "series": [
    { "name": "AirFare", "value": 108.42, "change_daily": 0.74 },
    { "name": "AirFare-uncorrected", "value": 111.07, "change_daily": 0.81 },
    { "name": "TCT-AirFare", "value": 114.23, "change_daily": 0.92 },
    { "name": "ANCx", "value": 121.55, "change_daily": 0.15 }
  ],
  "gaps": {
    "tct_minus_headline": 5.81,
    "correction_effect": -2.65
  },
  "coverage": 0.96, "provenance": "REAL",
  "methodology_version": "2.0.1"
}
```

11.4 GET /index/attribution/{date}

```
Path   date      ISO date
Query  series      default AirFare
Role   ANALYST

200
{
  "date": "2026-09-20", "series": "AirFare",
  "previous_value": 107.68, "current_value": 108.42,
  "total_movement": 0.74,
  "contributions": {
    "by_route": [ { "key": "DEL-BOM", "label": "Delhi - Mumbai",
      "points": 0.41, "share": 0.554 } ],
    "by_carrier": [ { "key": "6E", "label": "IndiGo",
      "points": 0.52, "share": 0.703 } ],
    "by_window": [ { "key": "T+1", "points": 0.33, "share": 0.446 } ],
    "by_component": [ { "key": "BASE_FARE", "points": 0.60 },
      { "key": "PLATFORM_FEE", "points": 0.09 } ],
    "by_driver": [ { "key": "FUEL", "points": 0.21 },
      { "key": "DEMAND", "points": 0.53 } ]
  },
  "reconciled": true
}
```

```
}
```

Each contribution set sums to total_movement. reconciled is the server's own assertion of that; the client re-checks it and renders a warning if either the flag is false or its own sum disagrees.

11.5 GET /routes

```
Query    date, sort (value|change|coverage|weight), order (asc|desc)
Role     VIEWER

200  envelope, data[]:
{
  "route_id": "DEL-BOM",
  "origin": "DEL", "destination": "BOM",
  "origin_state": "DL", "destination_state": "MH",
  "label": "Delhi - Mumbai",
  "weight_raw": 0.124, "weight_normalised": 0.3155,
  "index_value": 109.88, "change_daily": 0.62,
  "coverage": 0.98, "observation_count": 142,
  "sparkline": [106.1, 106.8, 107.2, 108.0, 109.9],
  "quality_status": "GOOD"
}
```

11.6 GET /routes/{routeId}/fares

```
Query    date, carrier, window, source
Role     VIEWER

200  envelope, data[]:
{
  "route_id": "DEL-BOM", "date": "2026-09-20",
  "advance_window": "T+7", "carrier": "6E", "source": "indigo",
  "cell_price_paise": 487500,
  "cell_price_display": "₹4,875",
  "observation_count": 8,
  "is_imputed": false, "imputation_rule": null,
  "is_low_confidence": false,
  "flags": []
}
```

11.7 GET /lead-time/matrix

```
Query    date, carrier (optional)
Role     VIEWER

200
{
  "date": "2026-09-20",
  "windows": ["T+1", "T+7", "T+15", "T+30", "T+45"],
  "scale": { "min_paise": 210000, "max_paise": 1480000 },
  "rows": [
    { "route_id": "DEL-BOM", "label": "Delhi - Mumbai",
      "cells": [
        { "window": "T+1", "price_paise": 1480000,
          "display": "₹14,800", "observation_count": 6,
          "state": "MEASURED" },
        { "window": "T+45", "price_paise": null,
          "display": null, "observation_count": 0,
          "state": "NO_DATA" }
      ]
    }
  ]
}
```

cell state is one of MEASURED, IMPUTED, LOW_CONFIDENCE or NO_DATA. The scale object is returned so the colour ramp is anchored consistently across dates; the client must not compute the scale from the visible data, or two dates stop being comparable.

11.8 GET /quality/coverage

```
Query    from, to
Role     VIEWER

200
{
  "date": "2026-09-20",
  "overall_score": 0.964,
  "components": {
    "coverage": { "value": 0.981, "target": 0.90, "status": "GOOD" },
    "completeness": { "value": 0.972, "target": 0.90, "status": "GOOD" },
    "source_agreement": { "value": 0.934, "target": 0.85, "status": "GOOD" },
    "freshness": { "value": 0.990, "target": 0.95, "status": "GOOD" },
    "validation": { "value": 0.948, "target": 0.95, "status": "WARNING" }
  },
  "cells_expected": 100, "cells_observed": 98,
  "provenance_mix": { "REAL": 0.40, "SIMULATED": 0.60 },
  "history": [ { "date": "2026-09-19", "overall_score": 0.958 } ]
}
```

11.9 GET /quality/imputation

```
200
{
  "date": "2026-09-20",
  "imputation_rate": 0.04,
  "by_rule": [
    { "rule": "CROSS_SOURCE", "count": 3, "share": 0.75 },
    { "rule": "CELL_MEAN", "count": 1, "share": 0.25 },
    { "rule": "CARRY_FORWARD", "count": 0, "share": 0.00 }
  ],
  "excluded_cells": 2,
  "carry_forward_at_cap": 0
}
```

11.10 GET /sources

```
200 envelope, data[]:
{
  "source": "indigo", "label": "IndiGo",
  "source_type": "AIRLINE", "tier": 1,
  "circuit_state": "HEALTHY",
  "success_rate_24h": 0.982,
  "last_success_at": "2026-09-21T02:31:44Z",
  "rate_limit_seconds": 3.5,
  "automation_allowed": true,
  "policy_last_reviewed": "2026-09-10",
  "adapter_version": "indigo:3.2.1"
}
```

11.11 GET /observations/{id}

Returns the full canonical observation as defined in TRD Section 15, with the raw artefact reference. This endpoint is what makes the provenance claim demonstrable, so it returns everything and hides nothing.

11.12 GET /methodology

```
200
{
  "methodology_version": "2.0.1",
  "base_period": "2026-09-01/2026-09-30",
  "elementary_aggregate": "GEOMETRIC_MEAN",
  "index_formula": "CHAINED_LASPEYRES",
  "chaining": "MONTHLY",
  "weights": [ { "route_id": "DEL-BOM", "raw": 0.124,
    "normalised": 0.3155, "source": "DGCA 2026-08" } ],
  "window_weights": [ { "window": "T+1", "weight": 0.10 } ],
  "models": {
    "hedonic": { "version": "h-1.2", "estimated_at": "2026-09-18" },
    "booking_curve": { "version": "bc-1.0",
      "calibration_source": "DGCA 2026-Q2" }
  }
}
```

```

    },
    "chain_drift": 0.14
  }

```

11.13 Remaining endpoints

Endpoint	Role	Note
GET /index/vintages/{date}	VIEWER	All vintages with prior value and revision reason
GET /index/diagnostics	ANALYST	Chain drift, formula cross-checks, active model versions
GET /states/{state}/index	VIEWER	State-level series for CPI integration
GET /carriers	VIEWER	Carrier reference and current aggregates
GET /components/breakdown	ANALYST	Fare component decomposition, aggregated
GET /ancillaries	ANALYST	Ancillary price observations
GET /export/csv	ANALYST	Streams CSV. Same filters as the source endpoint.
GET /export/sdmx	ANALYST	SDMX-ML for statistical exchange
POST /auth/login, /auth/refresh, /auth/logout	public	Session management

PART F · DATABASE

12. Schema

PostgreSQL 16 with TimescaleDB. Written so it can be typed directly into a migration. Enums first, because several tables depend on them.

12.1 Enums

```

CREATE TYPE source_type AS ENUM ('AIRLINE', 'OTA', 'REFERENCE');
CREATE TYPE fetch_tier AS ENUM ('HTTP', 'DYNAMIC', 'BROWSER');
CREATE TYPE circuit_state AS ENUM ('HEALTHY', 'DEGRADED', 'OPEN', 'RECOVERING');
CREATE TYPE missing_reason AS ENUM ('SOLD_OUT', 'NO_FLIGHT', 'MISSING_SOURCE',
    'SOURCE_ERROR', 'PARSER_ERROR', 'BLOCKED', 'UNKNOWN');
CREATE TYPE imputation_rule AS ENUM ('CROSS_SOURCE', 'CELL_MEAN',
    'CARRY_FORWARD', 'EXCLUDED');
CREATE TYPE provenance AS ENUM ('REAL', 'SIMULATED');
CREATE TYPE validation_status AS ENUM ('VALID', 'INVALID', 'FLAGGED');
CREATE TYPE job_state AS ENUM ('QUEUED', 'RUNNING', 'SUCCESS',
    'RETRY_WAIT', 'DEAD_LETTER');

```

12.2 Reference tables

```

CREATE TABLE sources (
  source text PRIMARY KEY,
  label text NOT NULL,
  type source_type NOT NULL,
  tier smallint NOT NULL,
  base_url text,
  adapter_version text NOT NULL,
  enabled boolean NOT NULL DEFAULT false
);

CREATE TABLE source_policies (
  source text PRIMARY KEY REFERENCES sources(source),
  robots_status text NOT NULL,
  automation_allowed boolean NOT NULL,
  rate_limit_seconds numeric(5,2) NOT NULL,
  approved_method fetch_tier NOT NULL,

```

```

    last_reviewed    date NOT NULL,
    owner            text NOT NULL,
    notes            text
);

CREATE TABLE routes (
    route_id         text PRIMARY KEY,
    origin            char(3) NOT NULL,
    destination       char(3) NOT NULL,
    origin_state      char(2) NOT NULL,
    destination_state char(2) NOT NULL,
    label             text NOT NULL,
    effective_from    date NOT NULL,
    effective_to       date
);

CREATE TABLE route_weights (
    id                bigserial PRIMARY KEY,
    route_id          text NOT NULL REFERENCES routes(route_id),
    weight_raw         numeric(8,6) NOT NULL,
    weight_normalised  numeric(8,6) NOT NULL,
    weight_version     text NOT NULL,
    weight_source      text NOT NULL,
    effective_from     date NOT NULL,
    effective_to       date,
    UNIQUE (route_id, weight_version)
);

CREATE TABLE carriers (
    carrier_code      char(2) PRIMARY KEY,
    label             text NOT NULL,
    is_lcc            boolean NOT NULL DEFAULT true
);

```

12.3 Collection tables

```

CREATE TABLE collection_runs (
    run_id            uuid PRIMARY KEY,
    started_at        timestampz NOT NULL,
    finished_at       timestampz,
    scheduled_slot    text NOT NULL,          -- 02:30 | 05:30 | 13:00 | 19:00
    jobs_total        integer NOT NULL DEFAULT 0,
    jobs_success      integer NOT NULL DEFAULT 0,
    jobs_failed       integer NOT NULL DEFAULT 0
);

CREATE TABLE scrape_jobs (
    job_id            uuid PRIMARY KEY,
    run_id            uuid NOT NULL REFERENCES collection_runs(run_id),
    source             text NOT NULL REFERENCES sources(source),
    route_id          text NOT NULL REFERENCES routes(route_id),
    departure_date     date NOT NULL,
    advance_window     smallint NOT NULL,
    state             job_state NOT NULL DEFAULT 'QUEUED',
    attempts           smallint NOT NULL DEFAULT 0,
    error_message      text,
    created_at         timestampz NOT NULL DEFAULT now(),
    UNIQUE (run_id, source, route_id, departure_date, advance_window)
);

CREATE TABLE raw_observations (
    raw_id            uuid PRIMARY KEY,
    run_id            uuid NOT NULL REFERENCES collection_runs(run_id),
    job_id            uuid NOT NULL REFERENCES scrape_jobs(job_id),
    source             text NOT NULL REFERENCES sources(source),
    object_key         text NOT NULL,          -- MinIO path
    sha256             char(64) NOT NULL,
    batch_hash         char(64) NOT NULL,      -- chained to previous batch
    prev_batch_hash    char(64),
    adapter_version    text NOT NULL,
    parser_version     text NOT NULL,
    fetched_at         timestampz NOT NULL
);
-- raw_observations is append-only. Enforce with a REVOKE on UPDATE/DELETE.

```

12.4 Observation tables

```
CREATE TABLE fare_observations (  
  observation_id      uuid NOT NULL,  
  observed_at         timestampz NOT NULL,  
  raw_id              uuid NOT NULL REFERENCES raw_observations(raw_id),  
  run_id              uuid NOT NULL,  
  source              text NOT NULL REFERENCES sources(source),  
  fetch_tier          fetch_tier NOT NULL,  
  route_id            text NOT NULL REFERENCES routes(route_id),  
  carrier_code        char(2) REFERENCES carriers(carrier_code),  
  flight_number       text,  
  departure_datetime  timestampz,  
  arrival_datetime    timestampz,  
  stops              smallint,  
  aircraft_type       text,  
  duration_minutes    integer,  
  search_date         date NOT NULL,  
  departure_date      date NOT NULL,  
  advance_days        smallint NOT NULL,  
  fare_family         text,  
  booking_class       text,  
  baggage_allowance_kg smallint,  
  refundable          boolean,  
  change_fee_paise    bigint,  
  cancellation_fee_paise bigint,  
  available           boolean NOT NULL DEFAULT true,  
  seats_remaining     smallint,  
  missing_reason       missing_reason,  
  validation_status    validation_status NOT NULL DEFAULT 'VALID',  
  component_completeness numeric(4,3),  
  provenance          provenance NOT NULL DEFAULT 'REAL',  
  fingerprint         char(64) NOT NULL,  
  PRIMARY KEY (observation_id, observed_at)  
);  
SELECT create_hypertable('fare_observations','observed_at');  
CREATE UNIQUE INDEX ux_fare_fingerprint  
  ON fare_observations (fingerprint, observed_at);  
CREATE INDEX ix_fare_cell  
  ON fare_observations (route_id, advance_days, observed_at DESC);  
CREATE INDEX ix_fare_source ON fare_observations (source, observed_at DESC);  
  
CREATE TABLE fare_components (  
  observation_id      uuid NOT NULL,  
  observed_at         timestampz NOT NULL,  
  base_fare_paise     bigint,  
  fuel_surcharge_paise bigint,  
  gst_paise           bigint,  
  udf_paise           bigint,  
  psf_paise           bigint,  
  platform_fee_paise  bigint,  
  other_mandatory_paise bigint,  
  total_payable_paise bigint NOT NULL,  
  currency            char(3) NOT NULL DEFAULT 'INR',  
  PRIMARY KEY (observation_id, observed_at)  
);  
  
CREATE TABLE ancillary_observations (  
  id                  bigserial PRIMARY KEY,  
  observation_id      uuid NOT NULL,  
  observed_at         timestampz NOT NULL,  
  type               text NOT NULL, -- CHECKED_BAG | SEAT | MEAL | ...  
  price_paise         bigint,  
  unit               text,  
  mandatory           boolean NOT NULL DEFAULT false  
);
```

Money is stored as integer paise throughout. A float rupee column will produce reconciliation failures in the arithmetic validation check and they will be intermittent, which is the worst kind of bug to find during a demonstration.

12.5 Statistics and index tables

```
CREATE TABLE quality_flags (  
  id          bigserial PRIMARY KEY,  
  observation_id uuid NOT NULL,  
  observed_at  timestampz NOT NULL,  
  flag        text NOT NULL,  -- OUTLIER_HIGH | LOW_CONFIDENCE | ...  
  detail      jsonb  
);  
  
CREATE TABLE imputations (  
  id          bigserial PRIMARY KEY,  
  index_date   date NOT NULL,  
  route_id    text NOT NULL,  
  advance_window smallint NOT NULL,  
  carrier_code char(2),  
  rule        imputation_rule NOT NULL,  
  imputed_value_paise bigint,  
  source_reason missing_reason,  
  carry_forward_days smallint DEFAULT 0,  
  UNIQUE (index_date, route_id, advance_window, carrier_code)  
);  
  
CREATE TABLE hedonic_model_runs (  
  model_version text PRIMARY KEY,  
  estimated_at  timestampz NOT NULL,  
  training_from date NOT NULL,  
  training_to   date NOT NULL,  
  observations_n integer NOT NULL,  
  r_squared     numeric(5,4),  
  coefficients  jsonb NOT NULL,  
  converged     boolean NOT NULL  
);  
  
CREATE TABLE booking_curves (  
  curve_version text NOT NULL,  
  route_id      text NOT NULL REFERENCES routes(route_id),  
  advance_window smallint NOT NULL,  
  booking_share numeric(5,4) NOT NULL,  
  calibration_source text NOT NULL,  
  calibrated_at  date NOT NULL,  
  PRIMARY KEY (curve_version, route_id, advance_window)  
);  
  
CREATE TABLE reference_series (  
  id          bigserial PRIMARY KEY,  
  series_name text NOT NULL,  -- DGCA_AVG_FARE | ATF_PRICE | USDINR  
  route_id    text,  
  period      date NOT NULL,  
  value       numeric(14,4) NOT NULL,  
  unit        text NOT NULL,  
  source_ref  text NOT NULL,  
  UNIQUE (series_name, route_id, period)  
);  
  
CREATE TABLE index_snapshots (  
  snapshot_id  uuid PRIMARY KEY,  
  index_name    text NOT NULL,  -- AirFare | TCT-AirFare | ANCx | ...  
  index_date    date NOT NULL,  
  value        numeric(10,4) NOT NULL,  
  base_value    numeric(10,4) NOT NULL DEFAULT 100,  
  base_period   text NOT NULL,  
  scope        text NOT NULL DEFAULT 'NATIONAL', -- or a state code  
  index_version text NOT NULL,  
  methodology_version text NOT NULL,  
  basket_version text NOT NULL,  
  weight_version text NOT NULL,  
  hedonic_model_version text,  
  booking_curve_version text,  
  observation_set_hash char(64) NOT NULL,  
  coverage_ratio numeric(4,3) NOT NULL,  
  imputation_ratio numeric(4,3) NOT NULL,  
  provenance    provenance NOT NULL,  
  vintage       smallint NOT NULL DEFAULT 1,  
  is_provisional boolean NOT NULL DEFAULT true,  
  calculated_at timestampz NOT NULL,  
  UNIQUE (index_name, index_date, scope, vintage)
```

```

);
SELECT create_hypertable('index_snapshots','index_date',
    chunk_time_interval => INTERVAL '1 month');

CREATE TABLE index_contributions (
    id                bigserial PRIMARY KEY,
    snapshot_id       uuid NOT NULL REFERENCES index_snapshots(snapshot_id),
    dimension          text NOT NULL, -- ROUTE|CARRIER|WINDOW|COMPONENT|DRIVER
    key               text NOT NULL,
    label             text,
    points            numeric(8,4) NOT NULL,
    share             numeric(6,4) NOT NULL,
    UNIQUE (snapshot_id, dimension, key)
);

CREATE TABLE index_revisions (
    id                bigserial PRIMARY KEY,
    index_name         text NOT NULL,
    index_date         date NOT NULL,
    prior_vintage      smallint NOT NULL,
    prior_value        numeric(10,4) NOT NULL,
    new_vintage        smallint NOT NULL,
    new_value          numeric(10,4) NOT NULL,
    reason             text NOT NULL,
    revised_at         timestampz NOT NULL DEFAULT now()
);

CREATE TABLE adapter_health (
    source             text NOT NULL REFERENCES sources(source),
    checked_at         timestampz NOT NULL,
    state              circuit_state NOT NULL,
    consecutive_failures smallint NOT NULL DEFAULT 0,
    success_rate_24h   numeric(4,3),
    last_success_at    timestampz,
    PRIMARY KEY (source, checked_at)
);

CREATE TABLE audit_events (
    id                bigserial PRIMARY KEY,
    occurred_at        timestampz NOT NULL DEFAULT now(),
    actor              text NOT NULL,
    action             text NOT NULL,
    entity             text NOT NULL,
    entity_id          text,
    detail             jsonb
);

```

12.6 Constraints worth enforcing in the database

- Normalised route weights for one weight_version must sum to 1. Enforce with a deferred constraint trigger, because a silent weight bug produces an index that looks plausible and is wrong.
- total_payable_paise must be greater than zero on any row with validation_status VALID.
- raw_observations has UPDATE and DELETE revoked from every application role.
- index_snapshots is unique on name, date, scope and vintage, so a recomputation must create a new vintage rather than overwrite.

PART G · CONFIGURATION & DATA

13. Environment Configuration

```

# Database
DATABASE_URL=postgresql://AirFare:<pw>@postgres:5432/AirFare
DB_POOL_MIN=2    DB_POOL_MAX=10

# Queue
REDIS_URL=redis://redis:6379/0
REDIS_STREAM_KEY=AirFare:collection
REDIS_CONSUMER_GROUP=collectors

```

```

# Object storage
S3_ENDPOINT=http://minio:9000
S3_BUCKET=AirFare-raw
S3_ACCESS_KEY=... S3_SECRET_KEY=...

# API
API_HOST=0.0.0.0 API_PORT=8000
JWT_SECRET=... JWT_EXPIRY_SECONDS=3600
REFRESH_EXPIRY_SECONDS=604800
CORS_ORIGINS=http://localhost:3000
RATE_LIMIT_PER_MINUTE=120

# Collection
COLLECTION_SLOTS=02:30,05:30,13:00,19:00
DEFAULT_RATE_LIMIT_SECONDS=3.5
MAX_RETRIES=4
CIRCUIT_FAILURE_THRESHOLD=10
USER_AGENT=AirFare-StatisticalCollector/2.0 (+https://<host>/AirFare-crawler)

# Statistics
BASE_PERIOD_START=2026-09-01 BASE_PERIOD_END=2026-09-30
METHODOLOGY_VERSION=2.0.1
OUTLIER_POOL_DAYS=14 OUTLIER_MIN_N=8
OUTLIER_MAD_THRESHOLD=3.5
CARRY_FORWARD_MAX_DAYS=3

# Frontend
NEXT_PUBLIC_API_BASE=http://localhost:8000/api/v1
NEXT_PUBLIC_USE MOCK=true

```

NEXT_PUBLIC_USE MOCK is the switch that lets the frontend be built before the backend exists. It must work on day one and must remain working, because it is also how the interface is developed after collection starts without depending on a live run.

13.1 Services and ports

Service	Port	Notes
dashboard	3000	Next.js
api	8000	FastAPI
postgres	5432	With the TimescaleDB extension. Named volume.
redis	6379	Streams. Append-only persistence enabled.
minio	9000 / 9001	API and console. Named volume.
scheduler	n/a	Airflow, or the cron fallback of TRD 12.3
worker	n/a	Scaled with docker compose up --scale worker=N
prometheus	9090	Optional for the demonstration
grafana	3001	Optional for the demonstration

13.2 First-run sequence

```

1 cp .env.example .env          and fill the secrets
2 docker compose up -d postgres redis minio
3 make migrate                  create schema and enums
4 make seed                     load reference + seed observations
5 docker compose up -d api dashboard
6 open http://localhost:3000    dashboard renders from seed data

```

Steps 1-6 must complete on a clean machine in under 10 minutes.
If they do not, fix that before writing any feature code.

14. Seed Data Specification

The seed dataset is not a convenience. It is what allows frontend and backend to be built in parallel, and it is what the demonstration falls back on if live collection fails on the day.

14.1 Required contents

Dataset	Specification
Reference data	5 routes with origin and destination states, 5 carriers, 8 sources with policy records, one weight version whose normalised weights sum to 1.
Observations	30 days x 5 routes x 5 windows x 4 sources, roughly 3,000 observations. Fare levels anchored to plausible DGCA route averages.
Realistic structure	T+1 fares 2.5x to 4x the T+45 fare. Weekend departures 15 to 25 per cent above weekday. One festival spike week. Intraday variance of 5 to 12 per cent within a day.
Component split	Base fare 60 to 70 per cent, fuel surcharge 8 to 12 per cent, UDF and PSF fixed per airport, GST 5 per cent, platform fee ₹350 to ₹550 on OTA sources only.
Ancillaries	Checked bag, seat and meal prices on 70 per cent of observations, null on the remainder so the unknown-segment rendering is exercised.
Deliberate gaps	Roughly 4 per cent of cells missing, spread across every missing_reason value, so every state has something to render.
Deliberate outliers	A handful of flagged outliers retained, so the flag-not-delete behaviour is visible.
Provenance split	First 18 days SIMULATED, last 12 REAL, with the boundary recorded so the chart rule has something to draw.
Index snapshots	One per day per series, with coverage, imputation ratio, vintage and provisional state populated. At least one date with two vintages so the revision interface is exercised.
Attribution	Contributions for every snapshot, summing exactly to the movement.
Health history	One source spending three days DEGRADED and one hour with the circuit OPEN, so those states render.

The rule for seed data

Seed data must exercise every state the interface can render, including the ugly ones. A seed set with perfect coverage and no gaps produces an interface that has never displayed a warning band, and the first time it does will be in front of judges.

14.2 Mock API

A mock server serves the contracts in Part E from static JSON derived from the seed set. It supports the same query parameters, returns the same envelope, and can be instructed to return errors and slow responses so loading and error states are testable. The frontend targets the mock until the real API passes contract tests, and the switch between them is one environment variable.

15. Acceptance

15.1 Per screen

A screen is done when every criterion in its Part C entry passes, plus all of the following.

- All five component states render correctly and can be triggered from the mock server.
- Any published value on the screen is accompanied by its quality metadata.
- The screen is usable by keyboard alone and every interactive element has a visible focus ring.
- Every chart has a working table-view toggle.
- The screen renders correctly in both light and dark mode, with dark values taken from the dark column of the palette rather than computed.
- The screen is complete at 768px width.
- No raw hex values appear in the component code. Tokens only.

15.2 Per endpoint

- Response matches the contract in Part E exactly, including field names and null behaviour.
- All documented error codes are reachable and return the standard error shape with a correlation id.
- Role enforcement is tested for each role, including the negative case.
- Any index-bearing response carries the full quality metadata block.
- A contract test runs in CI against the documented shape.

15.3 System level

#	Criterion
1	A clean machine reaches a rendering dashboard in under ten minutes using the first-run sequence.
2	The dashboard renders fully from seed data with the backend stopped and the mock switch on.
3	Every state in Section 5 can be produced on demand from the mock server.
4	The recompute command regenerates a published index value and reports a match.
5	Simulated data is visually distinguishable from real data on every chart that contains both.
6	Normalised route weights sum to 1, asserted both in the database and in the index engine.
7	No dual-axis chart exists anywhere in the application.
8	The methodology page is reachable without authentication.

16. Frontend Build Sequence

Order matters. This sequence keeps something demonstrable on screen from day two and front-loads the components everything else depends on.

Step	Build	Why here
1	Tokens, AppShell, nav, routing skeleton	Every page needs it. Half a day, unblocks everyone.
2	Mock server from the seed set	Removes the backend dependency entirely.

Step	Build	Why here
3	Card, StatTile, ChartFrame, DataTable, QualityBadge, ProvenanceTag	Six components cover roughly 80 per cent of the interface.
4	Overview page, end to end	The demonstration landing page. Existing early means it gets polished rather than rushed.
5	Lead-time heatmap	Highest visual impact per hour of work and the easiest thing for a judge to understand.
6	Attribution waterfall	The differentiator view. Harder, so it needs the time that steps 1 to 3 saved.
7	Cost comparison	Carries the core PRD evidence.
8	Routes list and detail with the observation drawer	Completes the drill-down and the provenance story.
9	Quality and Health	The trust group.
10	Carriers, Components, Methodology	Completeness.
11	Public view	Tier 2, only if time remains.

If the schedule slips

Steps 1 to 7 are the demonstration. Steps 8 to 11 are completeness. Cut from the bottom, never from the middle, and never cut the quality metadata from a screen to save time, because that is the one thing that distinguishes this interface from a flight-price dashboard.